Numerical Simulation of Quantized Vortices with Small Core Size via Vortex Tracking

Master-Seminar: Complex Quantum Systems and Vortex Dynamics

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1 Introduction

Ginzburg-Landau Hamiltonian:

$$E_\varepsilon(u) = \int_\Omega \frac{1}{2}|\nabla u|^2 + \frac{1}{4\varepsilon^2}(1-|u|^2)^2 \tag{1}$$

When investigating this variational model and minimizes with the singular limit $\varepsilon \to 0$.
The corresponding Schrödinger flow $i\partial_t\psi = -\nabla_{L^2}E_\varepsilon(\psi)$ On smooth bounded domain with Neumann boundary conditions one arrives at:

$$i\partial_t\psi_\varepsilon = \Delta\psi_\varepsilon(x,t) + \frac{1}{\varepsilon^2}(1-|\psi_\varepsilon(x,t)|^2)\psi_\varepsilon(x,t) \tag{2}$$

Observables:

- $|\psi|^2$
- Supercurrents $j(\psi) = \frac{1}{2i}(\overline{\psi}\nabla - \psi\nabla\overline{\psi})$
- Vorticity: $J\psi = \frac{1}{2}\nabla \times j(\psi)$

For vortices: $J\psi_\varepsilon(t) \to \pi\sum_{j=1}^N d_j\delta_{a_j(t)}$ The singular points solve:

$$\dot{a}_j(t) = -\frac{1}{\pi}d_j J\nabla_{a_j}W(a(t),d) \tag{3}$$

$$a_j(0) = a_j^0 \tag{4}$$

We introduce the harmonic map $u^*(a,d) \in W^{1,p}(\Omega,\mathbb{S}^{-1})$. What is it? Given vortex positions a and degrees d we want a map $u^* : \Omega \to \mathbb{C}$ that:

- lives on the unit circle away from vortex locations

- has exactly the right topological winding number d_j around each a_j
- satisfies homogeneous Neumann boundary conditions $\partial_\nu u^* = 0$ on $\partial\Omega$

The following conditions uniquely determine the supercurrents $j(u^*)$ which in turn determine u^*

$$\nabla j(u^*) = 0 \tag{5}$$

$$\nabla \times j(^*) = 2\pi \sum_{j=1}^{N} d_j \delta_{a_j} \tag{6}$$

$$\nu j(u^*) = 0 \text{ on } \partial\Omega \tag{7}$$

Where ν is the direction of the outward normal of $\partial\Omega$ u^* is then given by

$$u^*(x; a, d) = \exp iH(x) \prod_{j=1}^{N} \frac{x - a_j}{|x - a_j|}^{d_j} \tag{8}$$

Where $H : \mathbb{R}^2 \to \mathbb{R}^2$ is some harmonic function.
Boundary terms in the renormalized energy $R(x; a, d)$ which solves

$$\Delta R = 0, \text{ in } \Omega \tag{9}$$

$$R = -\sum_{j=1}^{N} d_j \log|x - a_j|, \text{ on } \partial\Omega \tag{10}$$

The supercurrents of u^* satisfies:

$$j(u^*) = -\nabla \times G, \quad G(x) = \sum_j d_j \log|x - a_j| + R(x; a, d) \tag{11}$$

This equation is important to compute the gradient of W u^* has singularities at the vortex locations a_j. This is the reason we need to smooth it out before using it as an initial condition for the PDE.
The renormalized energy: Why do we need it? The Ginzburg-Landau energy diverges as $\varepsilon \to 0$. This divergence comes purely from the self-energy of each vortex core and carries no information about vortex interactions. The renormalized energy is what remains after subtracting the divergence:

$$W(a, d) = -\pi \sum_{1 \le i \ne j \le N} d_i d_j \log|a_i - a_j - \pi \sum_j d_j R(a_j; a, d) \tag{12}$$

The first terms captures vortex-vortex interaction: same sign vortices repel, oppositve sign attract The second sum captures vortex-boundary interaction through R, the harmonic correction that enforces the Neumann boundary condition

Definition 1.1. A familiy of iniital conditions $(\psi_\varepsilon^0)_{\varepsilon>0}$ is siad to be well-prepared if it satisfies the following assumptions, for some constant $C >=; 0 < \alpha < 1$ and ε small enough:

1. There exists N vortices with positions $a^0 = (a_j^0)_{j=1,\ldots,N} \in \Omega^{*N}$ and degrees $d = (d_j)_{j=1,\ldots,N} \in \{\pm 1\}^N$ such that

$$||J\psi_\varepsilon^0 - \pi \sum_{j=1}^{N} d_j \delta_{a_j^0}|| \leq C\varepsilon^\alpha \tag{13}$$

   and the vortices are distant enough

2. the energy of ψ_ε^0 is close to be optimal:

$$E_\varepsilon(\psi_\varepsilon^0) \leq W_\varepsilon(a_0, d) + C\varepsilon^{1/2} \tag{14}$$

2 Method

2.1 Motivation

Why not just solve the PDE directly?

- The standard approach: time-splittinng + Fourier /FEM in space
- The core problem: resolving the vortex of size ε requires mesh size $h < \varepsilon$
- Adaptive FEM helps, but still expensive, the method in this paper shifts the bottleneck entirely

2.2 Algorithm

Step 1: Solve $1D$ ODE for f_ε compute initial phase H Step 2: Evolve vortex positions via Hamiltonian ODE (RK4, $\delta t = 10^{-3}$) Step 3: The radial prodfile f_ε solved once, scales with r_0/ε Key insight: ϵ only enters through the 1D preprocessing - it is no longer a discretization bottleneck

3 Error Analysis

Lemma 3.1. *Let $\Omega \subset \mathbb{R}^2$ be the unit disk, let $\{a(t)\}_{t\geq 0}$ denote the exact solution to the Hamiltonian mechanics. Let $b(t)$ be the approximated ODE trajectory with the same degrees and initial positions b^0 with harmonic polynomials of degree up to n and time step δ_t. Suppose that for some $T > 0$,*

$$\rho_T = \min_{t\leq T} \min_{j\neq k} \{|b_j(t) - b_k(t)|, dist(b_k(t), \partial\Omega), |a_j(t) - a_k(t)|, dist(a_j(t), \partial\Omega)\} > 0 \tag{15}$$

is such that $0 < \rho_T < 1$. *Then we have:*

$$|a(t) - b(t)| \leq C(|a^0 - b^0| + (1-\rho_T)^n n^{1-l}) + C\delta t^4 \tag{16}$$

In order to prove this we need 2 other lemmas

Lemma 3.2. *Let* $b^0, a^0 \in \Omega^{*N}$, *for fixed* $d \in \{\pm 1\}^N$ *and let* $\rho(a) := \min_{j \neq l}\{dist(a_j, \partial\Omega), |a_j - a_l|\}$. *Suppose that* $\rho_T := inf_{t\in[0,T]} \leq 0$

$$|a(t) - b(t)| \leq C(|a^0 - b^0| + \frac{sup_{s\in[0,T]}||\mathbb{P}^{|}_n g_{a(s)}||_{L^2(\partial\Omega)}}{\rho_T^{3/2}}) \exp Ct/\rho_T^{5/2} \tag{17}$$

Where $g_a(x) = -\sum_j^N d_j \log|x - a_j|$

The time integration errors follows from the previous bounds and standard considerations on the numerical integration of ODEs

Lemma 3.3. *Let* b *be the exact solution to* $\dot{b} = F^n(b)$ *and let* $(b^k_{\delta t})_{k\delta t \leq T}$ *be its numerical approximation with RK4 method and time step* δt

$$b^0_{\delta t} = b^0 \tag{18}$$

$$b^0_{\delta t} = b^{k-1}_{\delta t} + \delta t \Phi_{\delta t, n}(b^{k-1}_{\delta t}) \tag{19}$$

Where $\Phi_{\delta t,n}$ *is the RK4 increment function associated with the approximate forcing term* F^n. *Assume that:*

$$0 < \rho_T = \min(\inf_{0\leq t \ leqT} \rho(b(t)), \min_{k\delta t \leq T} \rho(b^k_{\delta t})) \tag{20}$$

Then there exists constants $C_{\rho_T} > 0$ *and* Λ_{ρ_T}, *that depends on* ρ_T *but not on* n, b *or* δt *such that the following in time error holds:*

$$\max_{k\delta t < T} |b(k\delta t) - b^k_{delta_t}| \leq \delta t^4 \frac{C_{\rho_T}}{\Lambda_{\rho_T}}(\exp(\Lambda_{\rho_T} T) - 1) \tag{21}$$